Steve Gottlieb
Going Deep

Seeking Interacting Galaxies

The Vorontsov-Velyaminov catalog is a mother lode of close galaxy pairs.

THE *PALOMAR OBSERVATORY Sky Survey*, conducted from 1949 to 1956 using the 48-inch Oschin Schmidt Telescope, recorded stars as faint as 21st magnitude, along with countless deep-sky objects. The 936 pairs of blue- and red-sensitive plates provided one of the most important resources for astronomical research for the next several decades.

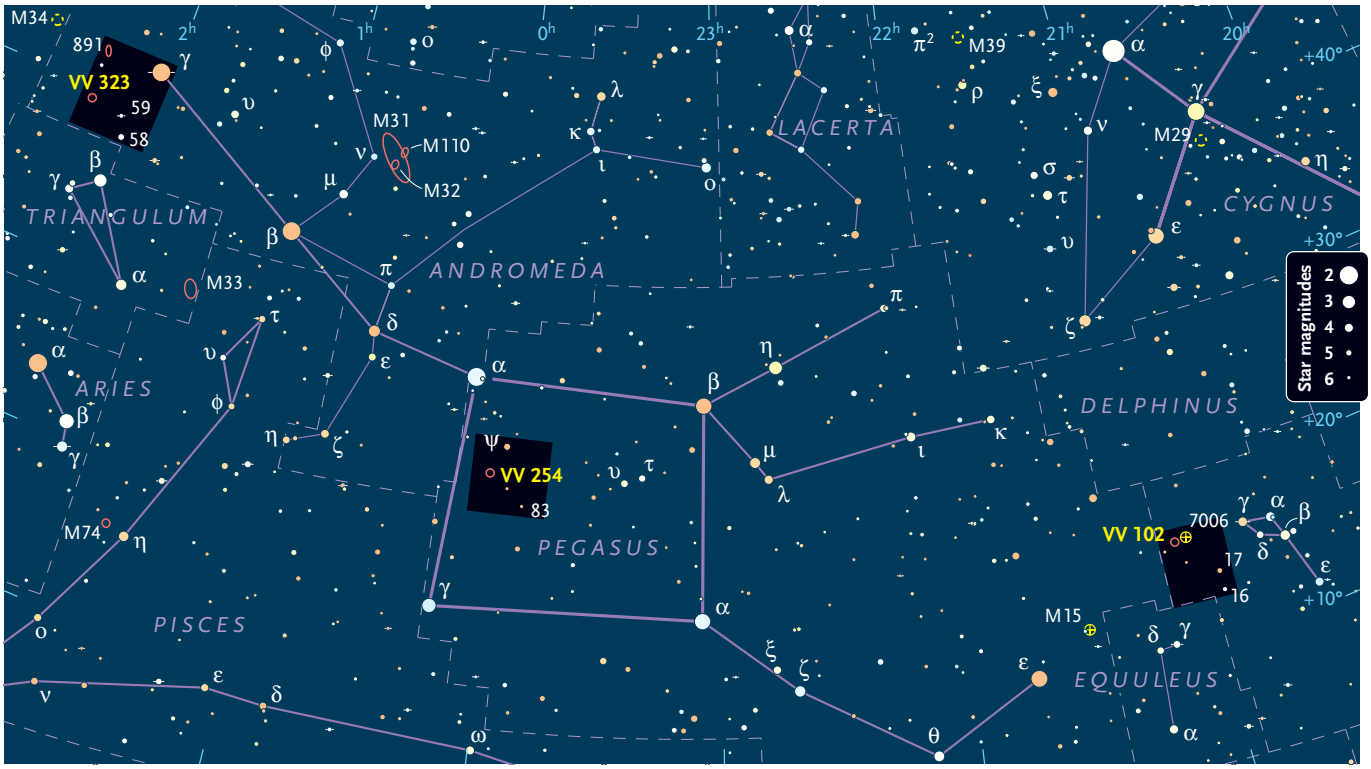
One of the first studies based on the new sky survey was the 1959 *Atlas and Catalogue of Interacting Galaxies* by Russian astronomer Boris Vorontsov-Velyaminov. It lists 355 systems with designations VV 1 through VV 355. Although some were previously known, Vorontsov-Velyaminov discovered most of them by visual inspection of the *Palomar Observatory Sky Survey* plates.

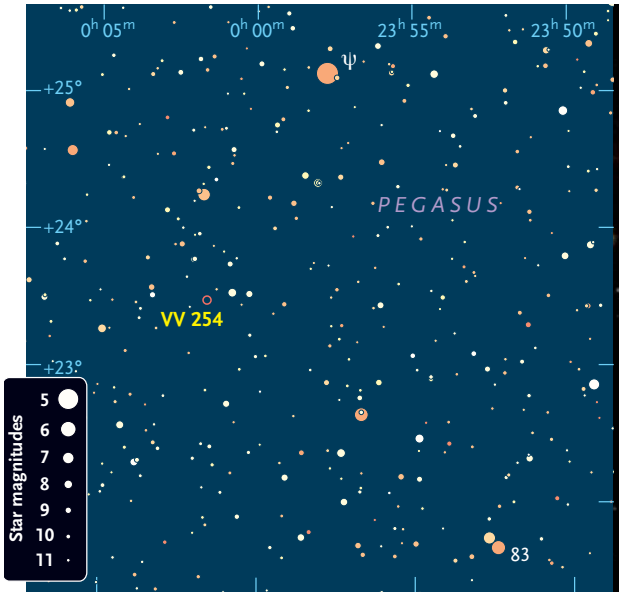
The first VV atlas divides interacting galaxies into 20 classes such as "merging or in contact," "nests of galaxies," and "chains with bridges." It also notes various interactions, such as tidal tails, distortions, and rings. The atlas showcases a number of bizarre, gravitationally deformed galaxies, and it caught the attention of the

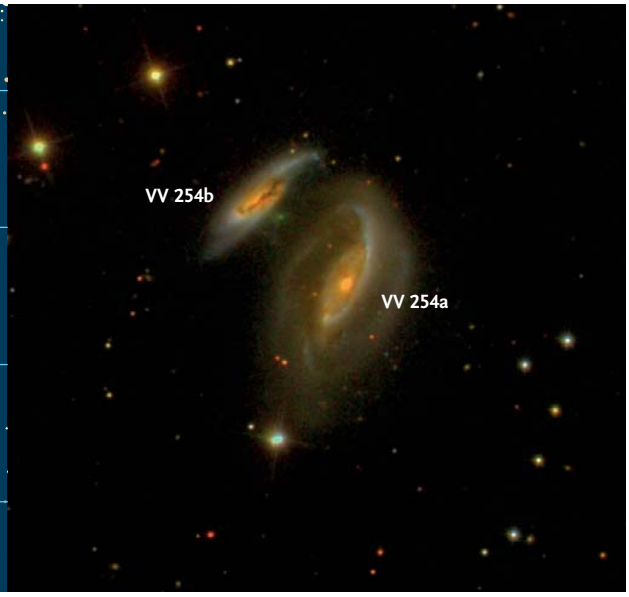
Selected Vorontsov-Velyaminov Galaxy Pairs

Name	UGC	Mag(s)	Const.	RA	Dec.
VV 254	12914, 12915	12.5, 13.0	Peg	0^h 01.6^m	+23° 29′
VV 323 (Arp 273)	1810, 1813	12.6, 14.2	And	2^h 21.5^m	+39° 23′
VV 102	11672	~14.5	Del	21^h 04.5^m	+16° 05′

Visual magnitudes have been estimated by correcting the blue magnitudes in the *Third Reference Catalog of Bright Galaxies* for galaxy type. Only a combined magnitude is available for VV 102; the components are too tightly bound to measure separately. Right ascension and declination are for equinox 2000.0.







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gifted, controversial astronomer Halton Arp. More than half of the 338 entries in Arp's 1966 *Atlas of Peculiar Galaxies* appeared earlier in the VV atlas.

In 1977 Vorontsov-Velyaminov published part II of his atlas. This supplement expanded the list of interacting galaxies to 852 and added new categories such as "galaxies of the M51 type," "chains in the making," and "blue nests." Finally, a 2001 update merged the first two catalogs and added 1,162 more interacting galaxies, bringing the designations up to VV 2014.

Although a 6- to 10-inch scope will reveal a number of the brighter VV systems, you'll probably need at least a 16-inch to view structural features of the interactions.

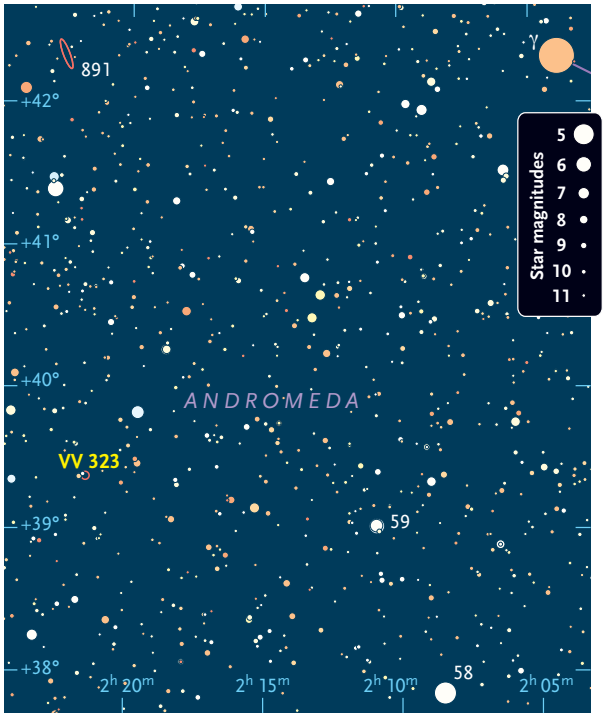
Heavenly Taffy

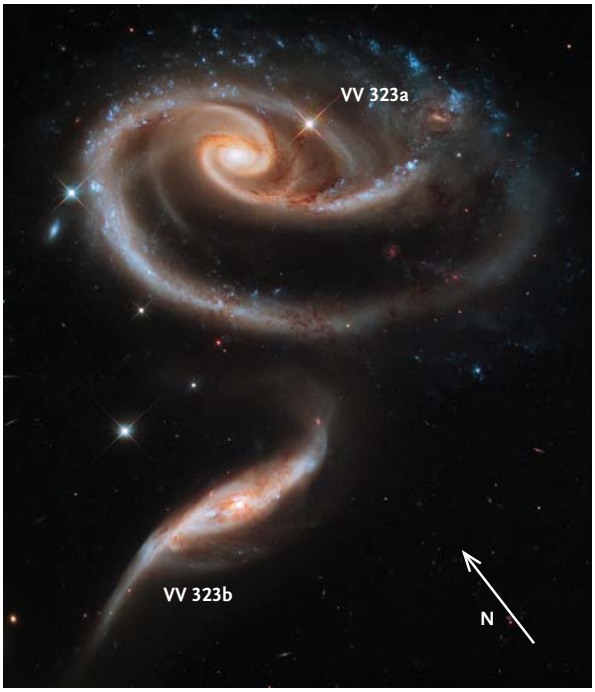
VV 254 is located along the eastern side of the Great Square of Pegasus, 5.8° south-southwest of 2.1-magnitude Alpheratz (Alpha Andromedae). It consists of two gas-rich spirals (UGC 12914 and 12915) receding from each other in the aftermath of a head-on collision that took place roughly 20 million years ago. In a 1993 study published in the *Astronomical Journal*, radio astronomer Jim Condon and colleagues coined the nickname Taffy Galaxies, because a radio-emission contour map of the gas bridge connecting the pair resembles strands of pulled taffy.

Through my 24-inch reflector at 375×, the Taffy Galaxies appear to be reaching out to each other. VV 254a, a 12.5-magnitude spiral stretching 1.6′ × 0.8′, is concentrated to a small, rounder core. A spiral arm emerges from the north-northwest end of the core and arcs toward VV 254b, just 1′ northeast. The 13th-magnitude companion spans 1.0′ x 0.4′ and also harbors a bright, compact core. From the northwest end of VV 254b, a short extension bends west toward VV 254a.

Hubble's Rose of Galaxies

Photogenic **VV 323** lies in easternmost Andromeda, 3.0° due south of the classic edge-on spiral galaxy NGC 891. Like the Taffy Galaxies, VV 323 is thought to be a post-collision result, with VV 323b (UGC 1813) piercing off-center through VV 323a (UGC 1810). The celestial mayhem disrupted VV 323a, resulting in highly warped spiral arms, a partial outer ring, and numerous blue knots of young, massive stars. Halton Arp included VV 323 in his category





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of "double galaxies with connected arms," so it also carries the designation Arp 273.

Both of VV 323's components show up faintly in my 18-inch reflector, but details are fleeting. Through Jimi Lowrey's 48-inch leviathan at 488×, 12.6-magnitude VV 323a has a chaotic, lopsided appearance. A prominent 15″ core condenses to a nearly stellar nucleus, and a 13.5-magnitude star is pinned to its west side. A narrow spiral arm emanates from the north side of the core and rotates counterclockwise east to a 14.5-magnitude star, then shoots south toward 14th-magnitude VV 323b, a 40″ × 10″ splinter trending east-west. Σ251, an attractive 2″-wide pair of 9th- and 10th-magnitude stars, lies 3′ east.

A Merging Duo

Vorontsov-Velyaminov placed **VV 102** (UGC 11672) in his interaction category of "coalescent pairs." It resides a mere 45′ east-southeast of the small globular cluster NGC 7006 in Delphinus and a similar distance west-southwest of a 12′ asterism that *Deep-Sky Wonders* columnist Sue French dubbed the "Toadstool."

The SDSS image below shows a fused double system with two nuclei separated by just 16″. At a redshift-based distance of roughly 420 million light-years, the twin nuclei might be as little as 35,000 light-years apart, depending how they're oriented to our line of sight.

Despite the tight separation, I could resolve the pair at 375× in my 24-inch. VV 102a, the slightly brighter eastern component, is a fairly faint 30″ irregular glow with a patchy surface brightness. VV 102b is a small 18″ knot jammed against the west side of VV 102a's halo.

You can find a scanned version of the original VV atlas at **ned.ipac.caltech.edu/level5/catalogs.html**. For a challenging observing project, Alvin Huey provides a downloadable guide at **www.faintfuzzies.com**. ✦

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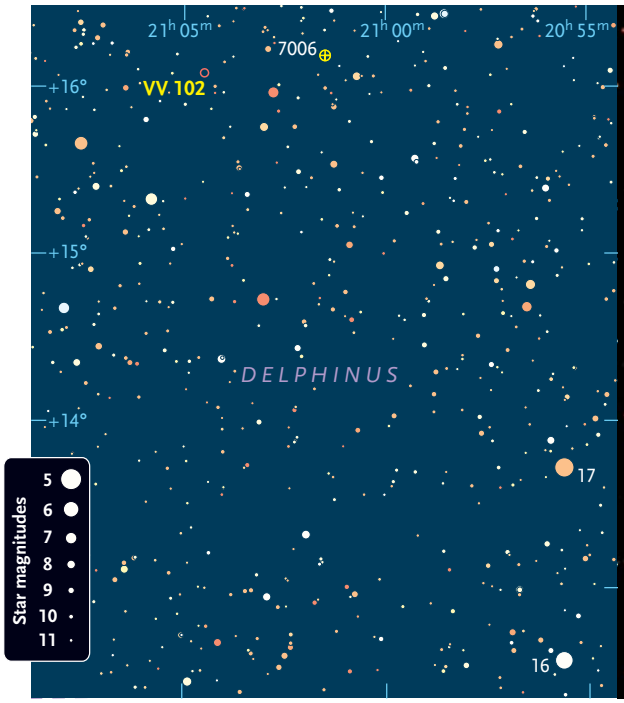
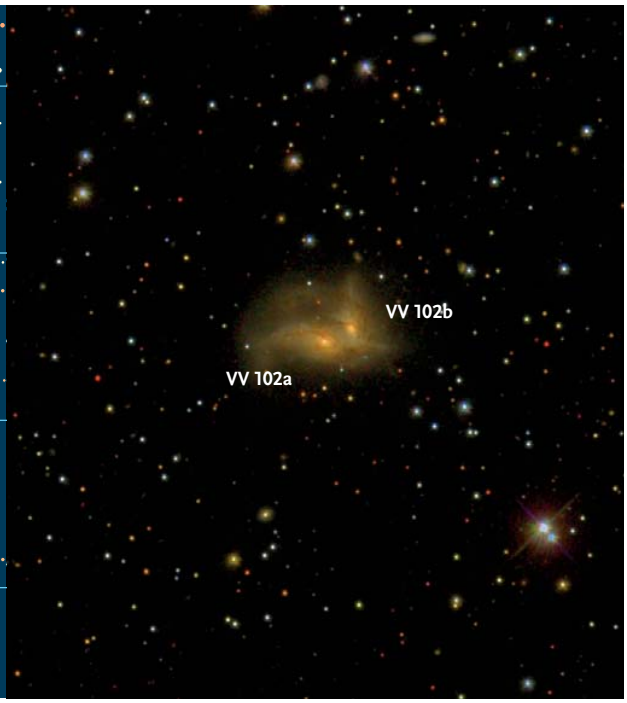
*Contributing editor **Steve Gottlieb** welcomes questions and comments at steve_gottlieb@comcast.net.*





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